46. If $|z|=1$ and $z \neq \pm 1$, then all the values of $\frac{z}{1-z^2}$ lie on

(A) a line not passing through the origin

(B) $|z|=\sqrt{2}$

(C) the x-axis

(D) the y-axis

Answer ○ (A) ○ (B) ○ (C) ● (D)

47. Let E^c denote the complement of an event E. Let E, F, G be pairwise independent events with $P(G)>0$ and $P(E \cap F \cap G)=0$. Then $P\left(E^c \cap F^c \mid G\right)$ equals

(A) $P\left(E^c\right)+P\left(F^c\right)$ (B) $P\left(E^c\right)-P\left(F^c\right)$

(C) $P\left(E^c\right)-P(F)$ (D) $P(E)-P\left(F^c\right)$

Answer ○ (A) ○ (B) ● (C) ○ (D)

48. $\frac{d^2x}{dy^2}$ equals

(A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

(C) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

Answer ○ (A) ○ (B) ○ (C) ● (D)

49. The differential equation $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{y}$ determines a family of circles with

(A) variable radii and a fixed centre at $(0, 1)$

(B) variable radii and a fixed centre at $(0, -1)$

(C) fixed radius 1 and variable centres along the x-axis

(D) fixed radius 1 and variable centres along the y-axis

Answer ○ (A) ○ (B) ● (C) ○ (D)

50. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Which one of the following is correct?

(A) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} = \vec{0}$

(B) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq \vec{0}$

(C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{a} \times \vec{c} \neq \vec{0}$

(D) $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ are mutually perpendicular

Answer ○ (A) ● (B) ○ (C) ○ (D)

51. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2\,CD$. Let AD be perpendicular to AB and CD. If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is

(A) 3 (B) 2 (C) $\frac{3}{2}$ (D) 1

Answer ○ (A) ● (B) ○ (C) ○ (D)

52. Let $f(x)=\dfrac{x}{\left(1+x^n\right)^{1/n}}$ for $n\geq 2$ and $g(x)=\underbrace{(f\circ f\circ\cdots\circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2}g(x)\,dx$ equals

(A) $\dfrac{1}{n(n-1)}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

(B) $\dfrac{1}{n-1}\left(1+nx^n\right)^{1-\frac{1}{n}}+K$

(C) $\dfrac{1}{n(n+1)}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

(D) $\dfrac{1}{n+1}\left(1+nx^n\right)^{1+\frac{1}{n}}+K$

Answer ● ○ ○ ○

(A) (B) (C) (D)

53. The letters of the word **COCHIN** are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word **COCHIN** is

(A) 360 (B) 192 (C) 96 (D) 48

Answer ○ ○ ● ○

(A) (B) (C) (D)

54. Consider the planes $3x-6y-2z=15$ and $2x+y-2z=5$.

STATEMENT-1 : The parametric equations of the line of intersection of the given planes are $x=3+14t,\ y=1+2t,\ z=15t$.

because

STATEMENT-2 : The vector $14\hat{i}+2\hat{j}+15\hat{k}$ is parallel to the line of intersection of given planes.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ○ ●

(A) (B) (C) (D)

55. STATEMENT-1 : The curve $y = \frac{-x^2}{2} + x + 1$ is symmetric with respect to the line $x = 1$.

because

STATEMENT-2 : A parabola is symmetric about its axis.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ● ○ ○ ○

(A) (B) (C) (D)

56. Let $f(x) = 2 + \cos x$ for all real x.

STATEMENT-1 : For each real t, there exists a point c in $[t, t+\pi]$ such that $f'(c) = 0$.

because

STATEMENT-2 : $f(t) = f(t + 2\pi)$ for each real t.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ● ○ ○

(A) (B) (C) (D)

57. Lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q, respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

STATEMENT-1 : The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$.

because

STATEMENT-2 : In any triangle, bisector of an angle divides the triangle into two similar triangles.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

58. Which one of the following statements is correct?

(A) $G_1 > G_2 > G_3 > \cdots$

(B) $G_1 < G_2 < G_3 < \cdots$

(C) $G_1 = G_2 = G_3 = \cdots$

(D) $G_1 < G_3 < G_5 < \cdots$ and $G_2 > G_4 > G_6 > \cdots$

Answer ○ ○ ● ○

(A) (B) (C) (D)

59. Which one of the following statements is correct?

(A) $A_1 > A_2 > A_3 > \cdots$

(B) $A_1 < A_2 < A_3 < \cdots$

(C) $A_1 > A_3 > A_5 > \cdots$ and $A_2 < A_4 < A_6 < \cdots$

(D) $A_1 < A_3 < A_5 < \cdots$ and $A_2 > A_4 > A_6 > \cdots$

Answer ● ○ ○ ○

(A) (B) (C) (D)

60. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > \cdots$

(B) $H_1 < H_2 < H_3 < \cdots$

(C) $H_1 > H_3 > H_5 > \cdots$ and $H_2 < H_4 < H_6 < \cdots$

(D) $H_1 < H_3 < H_5 < \cdots$ and $H_2 > H_4 > H_6 > \cdots$

M61-63 : Paragraph for Question Nos. 61 to 63

If a continuous function f defined on the real line **R**, assumes positive and negative values in **R** then the equation $f(x) = 0$ has a root in **R**. For example, if it is known that a continuous function f on **R** is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in **R**.

Consider $f(x) = ke^x - x$ for all real x where k is a real constant.

Answer ○ ● ○ ○

(A) (B) (C) (D)

61. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at

(A) no point (B) one point

(C) two points (D) more than two points

Answer ○ ● ○ ○

(A) (B) (C) (D)

62. The positive value of k for which $ke^x - x = 0$ has only one root is

(A) $\frac{1}{e}$ (B) 1 (C) e (D) $\log_e 2$

Answer ● ○ ○ ○

(A) (B) (C) (D)

63. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is

(A) $\left(0, \frac{1}{e}\right)$ (B) $\left(\frac{1}{e}, 1\right)$ (C) $\left(\frac{1}{e}, \infty\right)$ (D) $(0, 1)$

Answer ● ○ ○ ○

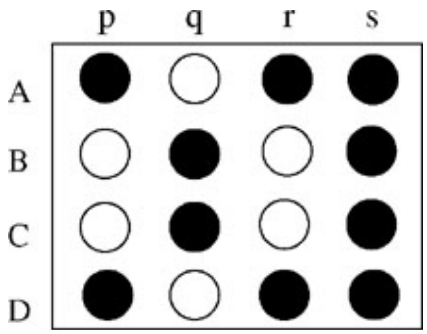
(A) (B) (C) (D)

64. Let $f(x)=\dfrac{x^2-6x+5}{x^2-5x+6}$.

Match the expressions/statements in **Column I** with expressions/statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) If $-1<x<1$, then $f(x)$ satisfies	(p) $0<f(x)<1$
(B) If $1<x<2$, then $f(x)$ satisfies	(q) $f(x)<0$
(C) If $3<x<5$, then $f(x)$ satisfies	(r) $f(x)>0$
(D) If $x>5$, then $f(x)$ satisfies	(s) $f(x)<1$

Answer



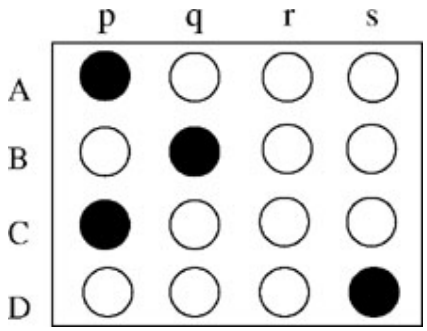
65. Let (x, y) be such that

$$\sin^{-1}(a\,x)+\cos^{-1}(y)+\cos^{-1}(b\,x\,y)=\frac{\pi}{2}.$$

Match the statements in **Column I** with statements in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in the ORS.

Column I	**Column II**
(A) If $a=1$ and $b=0$, then (x, y)	(p) lies on the circle $x^2+y^2=1$
(B) If $a=1$ and $b=1$, then (x, y)	(q) lies on $(x^2-1)(y^2-1)=0$
(C) If $a=1$ and $b=2$, then (x, y)	(r) lies on $y=x$
(D) If $a=2$ and $b=2$, then (x, y)	(s) lies on $(4x^2-1)(y^2-1)=0$

Answer



66. Match the statements in **Column I** with the properties in **Column II** and indicate your answer by darkening the appropriate bubbles in the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) Two intersecting circles	(p) have a common tangent
(B) Two mutually external circles	(q) have a common normal
(C) Two circles, one strictly inside the other	(r) do not have a common tangent
(D) Two branches of a hyperbola	(s) do not have a common normal

Answer

